

SOFTWARE REQUIREMENTS SPECIFICATION

for

AI-Based Question Paper Generator

Version 1.0 approved

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Revision History

Name	Date	Reason For Changes	Version
Ashrit Shukla	September 9, 2026	Initial Draft	1.0

1. Introduction

1.1 Purpose

The purpose of this Software Requirements Specification (SRS) document is to provide a detailed description of the **AI-Based Question Paper Generator** system. This document outlines the functional and non-functional requirements, system interfaces, and design constraints for the software.

The AI-Based Question Paper Generator is designed to automate the process of creating examination question papers using Artificial Intelligence and Natural Language Processing (NLP) technologies. The system allows educators to generate customized question papers based on subject matter, difficulty levels, Bloom's Taxonomy, and various question formats (MCQ, short answer, long answer, etc.).

1.2 Document Conventions

- **Bold text:** Indicates important terms, system features, or key concepts
- **Italic text**: Indicates user actions or specific values
- ``Monospace``: Indicates code, file names, or database fields
- **REQ-X**: Functional requirement identifier (e.g., REQ-1, REQ-2)
- **NFR-X**: Non-functional requirement identifier (e.g., NFR-1, NFR-2)
- Priority levels: **[High]**, **[Medium]**, **[Low]**

1.3 Intended Audience and Reading Suggestions

This document is intended for:

7. **Developers:** To understand system functionality and implement the software
8. **Project Managers:** To track project progress and scope
9. **Quality Assurance Team:** To create test cases and validate functionality
10. **End Users (Faculty/Educators):** To understand system capabilities
11. **Database Administrators:** To understand data storage requirements

Suggested Reading Sequence:

- Begin with Section 1 (Introduction) and Section 2 (Overall Description) for high-level understanding
- Section 3 (External Interface Requirements) for integration details
- Section 4 (System Features) for detailed functionality
- Section 5 (Nonfunctional Requirements) for quality standards

1.4 Product Scope

The AI-Based Question Paper Generator is a web-based application that leverages Artificial Intelligence (GPT models, NLP) to create examination papers automatically. The system addresses the following problems:

- **Manual Effort:** Eliminates hours of manual question paper creation
- **Consistency:** Ensures uniform quality and difficulty distribution
- **Customization:** Allows generation based on specific curriculum, board, university, or institutional patterns
- **Plagiarism Reduction:** Generates unique questions each time using AI
- **Time Efficiency:** Reduces paper creation time from days to minutes

Key Benefits:

- Auto-generation of questions from uploaded study material (PDF, DOCX, TXT)
- Intelligent difficulty level assignment (Easy, Medium, Hard)
- Bloom's Taxonomy classification (Remember, Understand, Apply, Analyze, Evaluate, Create)
- Multiple question formats: MCQ, True/False, Fill in the Blanks, Short Answer, Long Answer
- Customizable marking schemes and sections
- Export to PDF/Word format
- Question bank storage and management
- Duplicate question detection using similarity algorithms

Goals:

- Reduce question paper preparation time by 90%
- Ensure 100% coverage of specified topics/syllabus
- Maintain academic integrity with unique question generation
- Support multiple subjects, courses, and institutions

1.5 References

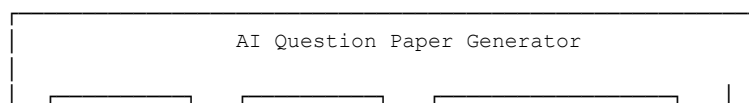
Document	**Description**
IEEE 830-1998	Recommended Practice for Software Requirements Specifications
OpenAI API Documentation	For GPT model integration
Bloom's Taxonomy Handbook	Taxonomy of Educational Objectives
KIET Group of Institutions Academic Guidelines	Question paper format standards

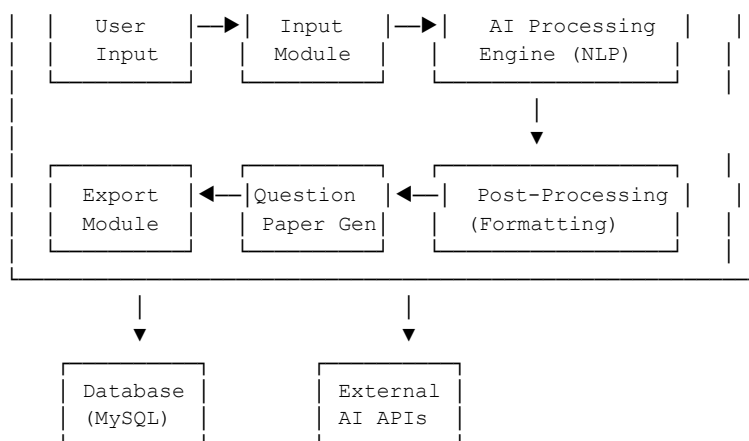
2. Overall Description

2.1 Product Perspective

The AI-Based Question Paper Generator is a **new, self-contained web-based system**. It is not a replacement for existing systems but introduces AI capabilities to the examination preparation workflow.

System Context Diagram: (Note: Set your Word font to Courier New for this block to display correctly)





2.2 Product Functions

The major functions of the AI-Based Question Paper Generator include:

12. **User Authentication:** Register, login, profile management
13. **Subject/Course Management:** Create and manage subjects, topics, and chapters
14. **Study Material Upload:** Upload PDFs, Word documents, or text files for question extraction
15. **Question Bank Management:** Store, edit, delete, and categorize generated questions
16. **AI Question Generation:** Generate questions automatically using NLP/AI models
17. **Question Paper Creation:** Create customized question papers with specific sections, marks, and difficulty levels
18. **Export & Download:** Export question papers to PDF, Word, or printable formats
19. **Duplicate Detection:** Avoid repeated questions using semantic similarity
20. **History & Analytics:** View previously generated papers and track usage

2.3 User Classes and Characteristics

User Class	**Description**	**Technical Expertise**	**Priority**
Faculty/Teacher	Primary user. Creates question papers for exams	Moderate (can use MS Word, Google Docs)	High
Administrator	Manages system, users, subjects, and AI model settings	High (understands system configuration)	High
Examination Committee	Reviews and approves generated papers	Low (only views/downloads papers)	Medium
Student	May access sample papers (if enabled)	Low	Low

2.4 Operating Environment

Hardware Requirements:

- **Server:** Minimum 4GB RAM, 2 CPU cores, 50GB SSD (for AI model hosting)
- **Client:** Any device with a modern web browser (Desktop/Laptop/Tablet)

Software Requirements:

- **Operating System:** Ubuntu 22.04 LTS (server), Any OS (client)
- **Web Server:** Nginx or Apache 2.4+

- **Database:** MySQL 8.0 or PostgreSQL 14+
- **Backend:** Python 3.9+ with Django/Flask framework
- **Frontend:** React.js 18+ or Vue.js 3+
- **AI Libraries:** TensorFlow/PyTorch, spaCy, Transformers (Hugging Face)
- **External APIs:** OpenAI GPT-4 API or local LLM (Llama, Mistral)

2.5 Design and Implementation Constraints

21. **AI Model Limitation:** GPT models may generate hallucinated or incorrect content. Human review required before finalizing papers.
22. **Data Privacy:** Study materials uploaded by faculty must not be shared with external services without consent.
23. **Scalability:** System should handle at least 100 concurrent users during examination season.
24. **Internet Dependency:** AI processing requires stable internet connection (if using cloud APIs).
25. **Language Support:** Primary support for English; Indian languages (Hindi, Marathi, Tamil) as optional enhancement.
26. **Cost Constraints:** Cloud AI API usage must be optimized to minimize operational costs.
27. **Regulatory Compliance:** Question papers must adhere to institutional/accreditation body formats (AICTE, UGC, etc.)

2.6 User Documentation

The following documentation will be provided:

28. **User Manual:** Step-by-step guide for faculty on creating question papers
29. **Administrator Guide:** System configuration, user management, AI model settings
30. **Video Tutorials:** Embedded help videos for first-time users
31. **FAQs:** Frequently asked questions
32. **API Documentation:** For developers integrating with external systems

2.7 Assumptions and Dependencies

Assumptions:

- Users have basic computer literacy and web browser access
- Institution has adequate hardware to run the system
- Users will review AI-generated content before finalizing papers
- Internet connectivity is available for AI processing
- Users will upload quality study materials for question generation

Dependencies:

- **OpenAI/LLM API:** System depends on external AI model availability
- **Database:** MySQL/PostgreSQL must be installed and configured
- **PDF Parser Library:** PyPDF2/PDF.js for extracting text from PDFs
- **Cloud Storage:** AWS S3/Google Cloud Storage for storing uploaded materials

3. External Interface Requirements

3.1 User Interfaces

Design Principles:

- Clean, intuitive, material-design-based UI
- Responsive design for mobile and desktop
- Dark/light theme toggle
- Hindi/English language toggle (future enhancement)

Screen Layouts:

Screen	**Description**	**Key Elements**
Login/Register	User authentication	Email, password, OTP, forgot password
Dashboard	Overview of recent papers, quick actions	Cards, charts, quick create button
Subject Management	Create/edit subjects and topics	Subject list, topic tree, add/edit/delete
Material Upload	Upload study material	Drag-drop zone, file list, progress bar
Question Generator	AI question generation settings	Subject, topic, difficulty, Bloom's level, question type
Paper Preview	Preview generated question paper	Live preview, edit individual questions, reorder
Export	Download question paper	PDF/Word format selection, layout options

3.2 Hardware Interfaces

The system is web-based and requires no specialized hardware. However, for local deployment:

- **Server:** Intel Xeon/AMD EPYC processor, minimum 8GB RAM
- **Storage:** 500GB HDD/SSD for question bank and study materials
- **Network:** 100Mbps ethernet connection for server

3.3 Software Interfaces

Interface	**Description**	**Protocol**
OpenAI API	For AI question generation using GPT-4	HTTPS REST API
MySQL Database	Storage of users, questions, papers	JDBC/ODBC
PDF Parser	Extract text from uploaded PDFs	PyPDF2 Library
Word Parser	Extract text from uploaded DOCX	python-docx
Email Service	For OTP, notifications	SMTP/IMAP
Storage Service	Store uploaded files	Amazon S3 / Local Filesystem

3.4 Communications Interfaces

- **Protocol:** HTTPS (secure), WebSocket (real-time updates for collaborative editing)
- **Data Format:** JSON for API responses, HTML/CSS/JavaScript for web interface
- **Authentication:** JWT (JSON Web Token), OAuth 2.0
- **File Upload:** Multipart/form-data

4. System Features

4.1 User Authentication and Management

4.1.1 Description and Priority

[High Priority] - Secure user registration, login, and role-based access control.

4.1.2 Stimulus/Response Sequences

- 33. User navigates to login page → System displays login form
- 34. User enters email and password → System validates credentials
- 35. User clicks "Forgot Password" → System sends OTP to email
- 36. User enters OTP → System verifies and allows password reset

4.1.3 Functional Requirements

ID	**Requirement**	**Priority**
REQ-1	System shall allow new user registration with name, email, department, and password	High
REQ-2	System shall send OTP for email verification during registration	High
REQ-3	System shall support login with email and password	High
REQ-4	System shall support password reset via email OTP	High
REQ-5	System shall enforce password policy (minimum 8 characters, alphanumeric, special character)	High
REQ-6	System shall support role-based access (Admin, Faculty, Reviewer)	High
REQ-7	System shall display error message for invalid credentials	Medium

4.2 Question Bank Management

4.2.1 Description and Priority

[High Priority] - Store, organize, and manage AI-generated and user-created questions.

4.2.2 Stimulus/Response Sequences

- 37. User views question bank → System displays list of questions with filters
- 38. User searches for questions → System filters based on subject, topic, difficulty
- 39. User edits a question → System saves changes and updates version
- 40. User deletes a question → System asks for confirmation then removes it

4.2.3 Functional Requirements

ID	**Requirement**	**Priority**
REQ-8	System shall store questions with attributes: subject, topic, difficulty, Bloom's level, marks, question type	High
REQ-9	System shall allow users to manually add questions to the bank	High

REQ-10	System shall support search and filter by subject, topic, difficulty, question type	High
REQ-11	System shall detect duplicate questions using semantic similarity (threshold > 85%)	High
REQ-12	System shall allow editing of existing questions	Medium
REQ-13	System shall allow deletion of questions with confirmation	Medium
REQ-14	System shall maintain question version history	Low

4.3 AI-Based Question Generation

4.3.1 Description and Priority

[High Priority] - Core feature that generates questions using AI/NLP models.

4.3.2 Stimulus/Response Sequences

41. User uploads study material → System extracts text and processes it
42. User selects topic and question type → System generates questions via AI
43. System displays generated questions → User reviews, edits, and accepts/rejects
44. User clicks "Generate More" → System generates additional unique questions

4.3.3 Functional Requirements

ID	**Requirement**	**Priority**
REQ-15	System shall generate questions from uploaded study material (PDF, DOCX, TXT)	High
REQ-16	System shall generate questions based on specified topic and subtopic	High
REQ-17	System shall support question types: MCQ, True/False, Fill in Blanks, Short Answer, Long Answer	High
REQ-18	System shall generate questions at specified difficulty levels (Easy, Medium, Hard)	High
REQ-19	System shall generate questions mapped to Bloom's Taxonomy levels	Medium
REQ-20	System shall generate a specified number of questions (e.g., "Generate 10 MCQs")	High
REQ-21	System shall allow users to regenerate questions if unsatisfied	Medium
REQ-22	System shall provide confidence score for each generated question	Low

4.4 Question Paper Generation

4.4.1 Description and Priority

[High Priority] - Create complete question papers with customizable sections, marks, and instructions.

4.4.2 Stimulus/Response Sequences

45. User selects "Create Question Paper" → System displays paper configuration form
 46. User sets subject, total marks, duration, number of sections → System validates input
 47. System auto-allocates questions based on difficulty distribution → User reviews allocation
 48. User clicks "Generate Paper" → System creates complete paper with instructions

4.4.3 Functional Requirements

ID	**Requirement**	**Priority**
REQ-23	System shall allow configuration of paper structure (sections, marks, time)	High
REQ-24	System shall support multiple question patterns (e.g., 10 MCQ × 1 mark, 5 Short × 3 marks)	High
REQ-25	System shall auto-select questions from bank based on difficulty distribution	High
REQ-26	System shall allow manual selection of specific questions	Medium
REQ-27	System shall calculate total marks and validate against configured total	High
REQ-28	System shall generate paper instructions (e.g., "Attempt all questions")	Medium
REQ-29	System shall support save as draft and finalize	High

4.5 Export and Download

4.5.1 Description and Priority

[High Priority] - Export generated question papers to PDF or Word format with proper formatting.

4.5.2 Stimulus/Response Sequences

49. User views paper preview → System displays formatted paper
 50. User clicks "Export PDF" → System generates PDF and downloads
 51. User clicks "Export Word" → System generates DOCX and downloads

4.5.3 Functional Requirements

ID	**Requirement**	**Priority**
REQ-30	System shall export question paper to PDF format	High
REQ-31	System shall export question paper to Word (DOCX) format	High
REQ-32	System shall allow selection of paper layout (A4, margins, font size)	Medium
REQ-33	System shall include institution name and logo in exported paper	Medium
REQ-34	System shall generate answer key for MCQ questions on export	Low

5. Other Nonfunctional Requirements

5.1 Performance Requirements

ID	Requirement
NFR-1	System shall generate 10 questions in less than 30 seconds
NFR-2	System shall load question bank with 10,000 questions in under 5 seconds
NFR-3	System shall export PDF in under 10 seconds
NFR-4	System shall support at least 100 concurrent users
NFR-5	System shall have 99.9% uptime during examination season

5.2 Security Requirements

ID	Requirement
NFR-6	All user passwords shall be hashed using bcrypt
NFR-7	All data transmission shall use HTTPS/SSL
NFR-8	System shall implement JWT token-based authentication
NFR-9	Study materials uploaded shall be accessible only to the owner
NFR-10	Question papers shall not be accessible to unauthorized users
NFR-11	System shall maintain audit log of all paper generation activities

5.3 Software Quality Attributes

Attribute	Description
Usability	System should be usable by faculty with minimal training (score > 80 on SUS)
Reliability	Mean Time Between Failures (MTBF) > 30 days
Maintainability	Modular architecture with documented code
Portability	Should work on Chrome, Firefox, Safari, Edge
Scalability	Should handle increase from 100 to 1000 users without code change

5.4 Business Rules

ID	Rule
BR-1	Only authenticated faculty can generate question papers
BR-2	AI-generated questions must be reviewed by faculty before final submission
BR-3	Maximum 100 questions per question paper
BR-4	Minimum 1 question per topic to ensure coverage
BR-5	MCQ questions default to 1 mark each unless changed

6. Other Requirements

52. **Database Requirements:** MySQL 8.0 with UTF-8 support for Indian languages

53. **Internationalization:** System should support English (primary) and Hindi (optional)

54. **Legal:** System must comply with institutional data privacy policies
55. **Backup:** Daily automated backup of question bank and user data
56. **Logging:** Maintain activity logs for 365 days

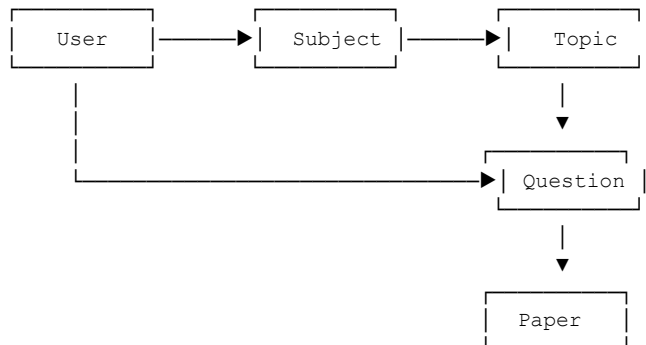
Appendix A: Glossary

Term	**Definition**
AI	Artificial Intelligence
NLP	Natural Language Processing
LLM	Large Language Model (GPT, Llama)
Bloom's Taxonomy	Classification of educational learning objectives
MCQ	Multiple Choice Question
SUS	System Usability Scale
JWT	JSON Web Token
OTP	One-Time Password
SRS	Software Requirements Specification

Appendix B: Analysis Models

Entity Relationship Diagram (High-Level)

(Note: Set your Word font to Courier New for this block to display correctly)*



Use Case Diagram (Text Representation)

Faculty:

- Login
- Upload Material
- Generate Questions
- Create Paper
- Export PDF
- Manage Question Bank

Administrator:

- Manage Users

- Manage Subjects
- Configure AI Settings
- View Analytics

Appendix C: To Be Determined List

ID	**Description**	**Status**
TBD-1	Exact AI model to be used (GPT-4 vs Open Source)	Pending cost analysis
TBD-2	Support for Indian languages (Hindi, Marathi)	Phase 2
TBD-3	Integration with Learning Management Systems (Moodle, Canvas)	Future
TBD-4	Multi-institution support	Future
TBD-5	Offline mode for AI generation	Research in progress

Are you ready to move on to designing the database schema or the user interface mockups for this project next?